

ECON 655: Experimental Economics

Field Experiments: An Introduction to Field Experiments

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1 Introduction

Field experimentation has been around for centuries. Researchers have used field experiments to identify and examine many questions throughout time. However, field experiments do not hold such longevity in the field of economics. Although in the early 20's and 30's there were field experiments being run, these experiments involved agriculture and land provision. However, economic field experiments have evolved and developed to look at economic phenomena. These phenomena include but are not limited to things such as individual behavior, biases, model and theory testing, and market predication. In the last decade economists have developed experiments using controls much like the lab. These experiments are broken into several categories: artefactual, framed and natural. I have selected four different experiments with a vast array of questions to explore more in-depth. I have chosen an experiment dealing with charitable fundraising, a experiment looking at punishment, fines and pricing, an individual and group decision making experiment, as well as an experiment looking at discrimination in the labor market. Although this is a small selection of the available literature I believe the chosen experiments portrays the versatility and benefits, as well as some possible downfalls of field experimentation in economics.

2 Field Experiments in Charitable Contribution: The Impact of Social Influence on the Voluntary Provision of Public Goods

Authors: Jen Shang and Rachel Croson

Publication: *The Economic Journal* 119:2009 (October) pages 1422-1439

The authors of this paper designed an experiment to look at social information on the voluntary provision of a public good. Throughout the literature there are theories that do not agree if other peoples contribution could be either substitutes or complements to one owns contribution. Therefore, the authors design an experiment to see the impact of this social information in a very interesting and unique way. They take an east coast radio station during an on-air fundraising drive and offer three treatments among random callers. They also allow themselves a control group that does not receive any treatment. This would allow the authors to expand on the existing literature in the areas of manipulating payoffs as well as social information on behavior. However, most field studies in this area have focuses on the content conveyed by the social information rather than the social influence component of the effect.

The design is set up so that people who are calling the radio station for this fundraising drive are given the treatment. Next the callers were greeted and given one of three treatments. The treatments varied in telling the caller how another member had contributed:

Treatment 1: \$75

Treatment 2: \$180

Treatment 3: \$300

The controls group was callers who received no knowledge of what others had contributed, this means that the control group did not receive any of the three treatments.

The authors also had to take into account that there were also thank you gifts given to the caller ranging from CD's and mugs to movie tickets and invitation to social events. These gifts were standard for the fundraising event and the station provided the gifts. The range of gifts depended on the amount of the contribution from the caller. The results that the authors found were compared in contrast with previous fund raising drives that occurred the same time of year, just in previous years.

The authors found that depending on the theme of the fund raiser, the thank you gifts offered, if the callers were new or renewing donors, the gender of the caller as well as if the contribution is made in one payment or in installments greatly affected the contributions. The authors also found that the mention of \$300 dollars increases contributions significantly more than the control. However, \$75 and \$180 are directional increasing, but are not significant. The authors state that if all the callers had been offered \$300 social information condition there would have been a 12% increase in revenue.

The Social Information Effect (standard errors in parentheses)

	All data		Without outliers	
	OLS	Robust regression	OLS	Robust regression
Constant	12.305 (69.283)	41.604+ (24.080)	35.967 (41.020)	41.389+ (23.839)
\$75	3.017 (13.337)	2.474 (4.635)	0.889 (7.972)	2.521 (4.633)
\$180	4.666 (11.215)	8.502* (3.898)	7.715 (6.674)	8.419* (3.879)
\$300	39.599** (13.609)	10.710* (4.730)	20.096* (8.126)	10.579* (4.722)
Renewing	36.405** (8.516)	9.956** (2.960)	15.319** (5.102)	9.923** (2.965)
Male	15.015+ (8.405)	0.813 (2.921)	11.558* (5.009)	0.789 (2.911)
Instalment	65.415** (8.634)	44.599** (2.960)	50.108** (5.164)	44.719** (3.001)
Drive	yes	yes	yes	yes
Day	yes	yes	yes	yes
Hour	yes	yes	yes	yes
N	538	538	530	530
R-Squared	0.180	0.366	0.232	0.370

**p < 0.01

*p < 0.05

+p < 0.10

Also the \$300 condition increased contribution in new members from approximately \$87 dollars to \$112. This is an increase in 29% of revenue. In a similar instance the returning callers contribution did not significantly increase.

The Social Information Effect is Significant for New Members but Not for Renewing Members

	New Members All data		New Members Without outliers		Renewing Members All Data		Renewing Members Without outliers	
	OLS	Robust Regression	OLS	Robust Regression	OLS	Robust Regression	OLS	Robust Regression
Constant	2.651 (57.858)	22.386 (23.910)	6.800 (42.824)	21.766 (23.601)	-43.169 (125.856)	45.931 (38.927)	56.649 (42.647)	57.176* (25.950)
\$75	4.769 (14.931)	0.445 (6.170)	6.518 (11.045)	0.613 (6.087)	-2.489 (24.729)	3.296 (7.649)	-13.946 (12.281)	2.735 (7.473)
\$180	12.636 (11.211)	11.744* (4.633)	15.450+ (8.297)	11.492* (4.572)	-5.643 (23.139)	0.978 (7.157)	-10.144 (11.470)	0.139 (6.941)
\$300	33.735* (13.531)	9.001+ (5.592)	23.373* (10.030)	8.458+ (5.528)	42.935 (30.010)	5.202 (9.282)	2.151 (15.017)	5.132 (9.138)
Male	14.406+ (8.590)	0.370 (3.550)	14.955* (6.354)	0.370 (3.502)	12.266 (17.033)	3.419 (5.268)	4.431 (8.430)	3.601 (5.130)
Instalment	56.469** (9.051)	50.354** (3.74)	50.949** (6.752)	50.705** (3.721)	83.298** (16.903)	41.452** (5.228)	53.491** (8.409)	40.978** (5.117)
Drive	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hour	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	317	317	315	315	221	221	215	215
R-Squared	0.194	0.466	0.269	0.476	0.230	0.309	0.284	0.317

**p < 0.01

*p < 0.05

+p < 0.10

The authors also look at the social conditions they implemented on renewal rates, and found that the renewal rate is higher in each case versus the control. They also found the contribution in the next year increases more than in the control. The final result in this analysis is that the expected revenue is also higher for the callers who had received one of the three treatments.

New Donors One Year Later

	Renewal rate (%)	Mean contribution (\$)	Expected revenue (\$)
Control	12	86.11	10.62
\$75	23	95.50	22.21
\$180	25	121.13	30.28
\$300	32	93.97	29.95

3 A Fine is a Price

Authors: Uri Gneezy and Also Rustichini

Publication: *Journal of Legal Studies*, 29(1):1-17

The authors design an experiment to look at the deterrence hypothesis. The deterrence hypothesis predicts that implementing a fine or penalty will reduce the behavior that is trying to be changed, if all else stays the same. The authors examine this by implementing a field study on a group of day-care centers in Israel.

The field study was implemented in private day-care centers in the Israel city of Haifa. All the daycare centers were similarly located and over all very similar priced. The other important factor was that each daycare had similar number of children enrolled. These day-care centers offer a rich resource to look at the implementation of fines on parents that are picking up their children late.

The authors designed a field study where they asked the principals of these centers to participate in an academic study about the implementation of fines. The authors offered a monetary compensation to the day-care centers of a book allotment for participation. The experiment would also offer additional revenue through fines from parents picking up late students.

The study was a 20 week study where the first four weeks the authors just observed the number of parents who arrived late in picking up their children. In the fifth week the authors implemented a fine to parents who decided to arrive late in picking up their children at six out of a total of ten day-care centers in the study. The centers were chosen at random for the fine implementation. The other day care centers were selected to be the control group, not receiving a fine.

This fine was announced through a consistent method of placing a note on the bulletin board where important announcements were normally located. The fine that was to be implemented was for any time that the parents were late more than ten minutes they received a NIS 10 fine. This fine was per child as well as per occurrence. The money was paid to the principal of the center at the end of the month. However, the principal was not the individual that was watching the children if their parents were to arrive late, the day-care workers were the ones overseeing this role. Finally the study concluded with the fine being removed without any explanation other than a notice placed on the same bulletin board. Then the experimenters observed the parent's behavior after the fine had been removed for three more weeks.

The result that the authors found was not what was predicted by the deterrence hypothesis. As soon as the fine was implemented for late arriving parents, the number of parents that came late increased significantly even after the fine had been removed toward the end of the study.

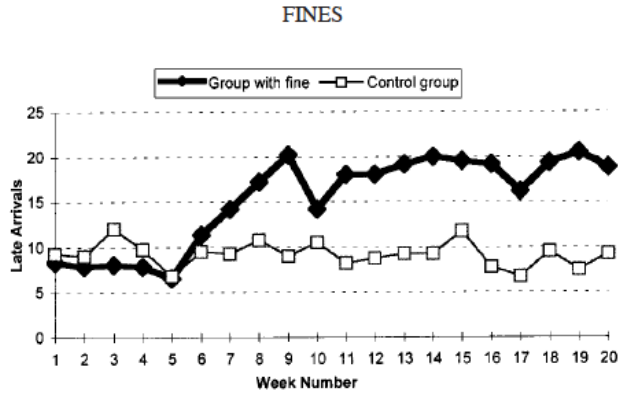


FIGURE 1.—Average number of late-coming parents, per week

The authors cleverly designed this experiment so that nothing else would be changed other than the implementation of the fine itself. Since the authors only implemented the fine at 60% of the centers, it allowed the other 40% to be used as a control group to compare the late-arriving parents to. The rate of late arriving parents was almost twice as high after the fine was implemented than to the previous arrangement. Another interesting finding that the authors were able to examine was that even after the fine was removed the number of late arrivals remained unchanged and unaffected.

TABLE 2
AVERAGE NUMBER OF LATE-COMING PARENTS, ACCORDING TO
FOUR PERIODS OF THE STUDY

Center	No. of Children	Weeks 1–4	Weeks 5–8	Weeks 5–16	Weeks 17–20
1	37	7.25	9.5	12.5	15.25
2	35	5.25	9	12.2	13.25
3	35	8.5	10.25	16.8	22
4	34	9	15	19.1	20.25
5	33	11.75	20	24.6	29.5
6	28	6.25	10	13.1	12
7	35	8.75	8	7.2	6.75
8	34	13.25	10.5	10.9	9.25
9	34	4.75	5.5	5.5	4.75
10	32	13.25	12.25	13.1	12.25

NOTE.—The four periods of the study are as follows: before the fine (weeks 1–4), the first 4 weeks with the fine (weeks 5–8), the entire period with the fine (weeks 5–16), and the postfine period (weeks 17–20).

These findings of the increase late arrivals do not fit in with the original hypothesis and the previous literature did not provide an explanation as to why this would be happening. The authors then build a model trying to explain how implementing the fine changed the parent's perception on arriving late. The model that the authors offer here although complicated allow for the behavior that was recorded in the study they designed. This model has three equilibrium predictions due to the

possibility of two different types of principals at the day-care centers. There is multiple equilibrium as well as social norm factors that play an important role in parent's decisions to pick their child up late.

The authors also design another small experiment to look at one specific social norm that "Once a commodity, always a commodity". They did this by looking at rewards based on performance. They had four groups solve 50 questions from an IQ test and paid all four groups a flat fee of NIS 60, while offer additional money to three of the four groups for each correct answer. The amounts varied from NIS .10, NIS 1, and NIS 3. The fourth group had no additional monies for correct answers. The NIS.10 group did worse than the group offered nothing else in addition. The authors conclude from this that the "absence of a price is not just the limit of very low prices." However mentioning a price makes it go from a service to a market exchange. Concluding that the fine they implemented in the day care centers is enough to change the perception that they are obligated to arrive for pick-up on time.

3 Are Emily and Greg More Employable than Lakisha and Jamal?

A Field Experiment on Labor Market Discrimination

Authors: Marianne Bertrand and Sendhil Mullainathan

Publication: *The American Economic Review*, Vol. 94, No. 4 (Sep., 2004), pp.

991-1013

The authors design a study to look at the effects of a name on possible employment. Although this design involves deception, and has been criticized as creating more work for human resource employees, it offers a different approach to solving an economic problem with field experimentation. The design implementation allows the authors to tease out the discrimination aspect through random name assignment. This scope of field experiments offers a glimpse at how fields like labor economics can benefit from the data gathered from field experiments.

The authors created a bank of resumes, leaving the important feature of the name off as well as varying the quality. They create these resumes from manipulating publically available resumes from online job search websites. They restricted the job market to Boston and Chicago, as well as the field of jobs to sales, administrative support, clerical service and customer service.

They use the bank of resumes as well as a list of names that have been decided to be uniquely African American and uniquely white. The resumes are then sent out to jobs that advertise in the major newspaper for each selected area. The resumes are then blocked in this order:

Uniquely African-American Name:

High Quality

Low Quality

Uniquely White Name:

High Quality

Low Quality

The authors then monitor if the resumes receive a call back or an email. The authors were unable to monitor mail solicitations, due to fictitious addresses used on the resumes. The results the authors find shock many people. The white sounding names were 49-50% more likely to receive a call back than the resumes that bore an African-American name. They do abundant analysis on the data and find that this result is for both markets, as well as both qualities of resumes.

TABLE 1—MEAN CALLBACK RATES BY RACIAL SOUNDINGNESS OF NAMES

	Percent callback for White names	Percent callback for African-American names	Ratio	Percent difference (<i>p</i> -value)
Sample:				
All sent resumes	9.65 [2,435]	6.45 [2,435]	1.50	3.20 (0.0000)
Chicago	8.06 [1,352]	5.40 [1,352]	1.49	2.66 (0.0057)
Boston	11.63 [1,083]	7.76 [1,083]	1.50	4.05 (0.0023)
Females	9.89 [1,860]	6.63 [1,886]	1.49	3.26 (0.0003)
Females in administrative jobs	10.46 [1,358]	6.55 [1,359]	1.60	3.91 (0.0003)
Females in sales jobs	8.37 [502]	6.83 [527]	1.22	1.54 (0.3523)
Males	8.87 [575]	5.83 [549]	1.52	3.04 (0.0513)

Notes: The table reports, for the entire sample and different subsamples of sent resumes, the callback rates for applicants with a White-sounding name (column 1) an an African-American-sounding name (column 2), as well as the ratio (column 3) and difference (column 4) of these callback rates. In brackets in each cell is the number of resumes sent in that cell. Column 4 also reports the *p*-value for a test of proportion testing the null hypothesis that the callback rates are equal across racial groups.

The authors also looked at the name and if it held different returns to resume quality. They find that the returns for African-American sounding names were statistically weaker than those resumes with white names. The quality of the resume varied through years of experience, a volunteering factor, military experience, and holes in employment, schooling, as well as individual skills.

TABLE 5—EFFECT OF RESUME CHARACTERISTICS ON LIKELIHOOD OF CALLBACK

Dependent Variable: Callback Dummy Sample:	All resumes	White names	African-American names
Years of experience (*10)	0.07 (0.03)	0.13 (0.04)	0.02 (0.03)
Years of experience ² (*100)	-0.02 (0.01)	-0.04 (0.01)	-0.00 (0.01)
Volunteering? (Y = 1)	-0.01 (0.01)	-0.01 (0.01)	0.01 (0.01)
Military experience? (Y = 1)	-0.00 (0.01)	0.02 (0.03)	-0.01 (0.02)
E-mail? (Y = 1)	0.02 (0.01)	0.03 (0.01)	-0.00 (0.01)
Employment holes? (Y = 1)	0.02 (0.01)	0.03 (0.02)	0.01 (0.01)
Work in school? (Y = 1)	0.01 (0.01)	0.02 (0.01)	-0.00 (0.01)
Honors? (Y = 1)	0.05 (0.02)	0.06 (0.03)	0.03 (0.02)
Computer skills? (Y = 1)	-0.02 (0.01)	-0.04 (0.02)	-0.00 (0.01)
Special skills? (Y = 1)	0.05 (0.01)	0.06 (0.02)	0.04 (0.01)
<i>H₀</i> : Resume characteristics effects are all zero (<i>p</i> -value)	54.50 (0.0000)	57.59 (0.0000)	23.85 (0.0080)
Standard deviation of predicted callback	0.047	0.062	0.037
Sample size	4,870	2,435	2,435

Notes: Each column gives the results of a probit regression where the dependent variable is the callback dummy. Reported in the table are estimated marginal changes in probability for the continuous variables and estimated discrete changes for the dummy variables. Also included in each regression are a city dummy, a sex dummy, six occupation dummies, and a vector of dummy variables for job requirements as listed in the employment ad (see Section III, subsection D, for details). Sample in column 1 is the entire set of sent resumes; sample in column 2 is the set of resumes with White names; sample in column 3 is the set of resumes with African-American names. Standard errors are corrected for clustering of the observations at the employment-ad level. Reported in the second to last row are the *p*-values for a χ^2 testing that the effects on the resume characteristics are all zero. Reported in the second to last row is the standard deviation of the predicted callback rate.

4 In Search of Homo Economicus: Behavioral Experiments in 15 Small-Scale Societies

Authors: Joseph Henrich, Robert Boyd, Samuel Bowles, Colin Camerer, Ernst Fehr, Herbert Gintis and Richard McElreath

Publication: *The American Economic Review*, 91(2): 73-78

The authors of this publication used a study that examined behavior subjects exhibited in the ultimatum game. It took the ultimatum game that has been run all over the world at universities and took it to the field. The authors examined subject's behavior in the ultimatum game in 15 small-scale societies. This work was part of a larger social study that resulted in these ultimatum games being played around the world. There were many researchers involved including economist and anthropologist. The first ultimatum game was played with the Machiguenga. The overall social experiment included 3 hunter-gather societies, 6 horticulturalist societies, 4 transhumant/nomadic herders, and 4 small-scale sedentary farmers. The authors played ultimatum, public goods, and dictator and trust games. The authors find that the behavior varies depending on the cultural and tribal norms of the small societies. The group differences they find are strikingly different

compared to previous work.

Group	Country	Mean Offer	Modes (% of sample)	Rejection Rate	Rejections 20% of pot
Machiguenga	Peru	0.26	0.15/0.25 (72%)	1/21	1/10
Hadza (Small Camp)	Tanzania	0.27	0.20 (38%)	8/29	5/16
Tsimané	Bolivia	0.37	0.5/0.3/0.25	0/70	0/5
Quichua	Ecuador	0.27	0.25 (47%)	2/13	1/2
Hadza (all camps)	Tanzania	0.33	0.20/0.50 (47%)	13/55	9/21
Torguud	Mongolia	0.35	0.25 (30%)	1/20	0/1
Khazax	Mongolia	0.36	0.25		
Mapuche	Chile	0.34	0.50/0.33 (46%)	2/30	2/10
Au	PNG	0.43	0.3 (33%)	8/30	1/1
Gnau	PNG	0.38	0.4 (32%)	10/25	3/6
Hadza (Big Camp)	Tanzania	0.40	0.50 (28%)	5/26	4/5
Sangu (farmers)	Tanzania	0.41	0.50 (35%)	5/20	1/1
Unresettled	Zimbabwe	0.41	0.50 (56%)	3/31	2/5
Achuar	Ecuador	0.42	0.50 (36%)	0/16	0/1
Sangu (herders)	Tanzania	0.42	0.50 (40%)	1/20	1/1
Orma	Kenya	0.44	0.50 (54%)	2/56	0/0
Resettled	Zimbabwe	0.45	0.50 (70%)	12/86	4/7
Ache	Paraguay	0.51	0.50/0.40 (75%)	0/5	0/8
Lamelara	Indonesia	0.58	0.50 (63%)	0/2	0.37

Typical demographic information such as age, sex, and wealth do not explain this difference however there are a few exceptions. Overall, these demographic variables are not an indicator on behavior in the ultimatum game. The authors also discover that education, wealth (described as ownership of animals, cash, crops or land), and the wage for labor also offer no explanatory power. The next things the authors look to for explanation include risk preference, language competence, math skill and pure understanding offer no decisive answer to the phenomena. The authors offer explanation through several rank variables including payoffs to cooperation and market integration.

Examining the dictator game, and other simple games in the 15 small-scale societies, allow researchers the capability of identifying drivers and the difference in the drivers through various societies. This small piece to a large social project offer data to solidify behavior of subjects from an ultimatum game in the laboratory, can and often are extremely different than those that are found in rural small-scale tribal communities.

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